

Exercise 5 – Analysing and aligning newly discovered proteins

Objectives:

- to apply what has been learned today

Anonymous Test Proteins:

below, we provide 20 randomly chosen proteins. All have been derived from DNA on the teeth of ancient skeletons found in a german monastery (same as for the previous exercises). None of the proteins have been analyzed in detail before ... Please select arbitrarily one of the proteins below, and analyze it like we did in exercises #1 through #3.

Optionally you can also study some protein sequence related to SARS-CoV-2, which are listed at the bottom below.

Questions:

- what protein family does your protein belong to?
- which domain(s), if any, does the protein contain?
- from which organism is it, likely?
- what function might it have?
- is it complete?
- how can it be best aligned to other members of its family?

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>NODE_4178_length_1047_cov_6.240688_S6
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 GAFDQAEALAEYRSFFKDAPFVTVLPEGRQPRTVSVAGTNYAHVSACYNERAGAVVATCA
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SARS-CoV-2 related proteins:

>pdb|6YLA|A Chain A, SARS-CoV-2 RBD
 ETGPNITNLCPFGEVFNATRFASVYAWNKRKISNCVADYSVLYNSASFSTFKCYGVSPTKLNDLCFTNVY
 ADSFVIRGDEVQRQIAPGQTGKIADYNYKL PDDFTGCVIAWNSNNLDSKVGGNYNLYRLFRKSNLKPFFER
 DISTEIIYQAGSTPCNGVEGFNCYFPLQSYGFQPTNGVGYQPYRVVVL SFELLHAPATVCGPKKSTNKH
 HHH

Or go to this URL:

<https://tinyurl.com/2f8h5vwf>

(https://www.ncbi.nlm.nih.gov/protein/?term=Severe+acute+respiratory+syndrome+coronavirus+2%5Borganism%5D+AND+protein_structure_direct%5Bfilt%5D)

click any SARS-CoV-2 related protein name and then click “FASTA” button at the top of the new page and use them to repeat exercise 1-3